

Appendix F: Classical Relativity as Macro-Geometry

F.1 Gauss-Codazzi Projection of the 6D Einstein-Hilbert Action

The BMI framework operates on a 6D pseudo-Riemannian manifold. The 6D Einstein-Hilbert action is:

$$S_{EH} = (1 / 2\kappa_6^2) \int d^6x \sqrt{(-G)} R_6$$

Using the Gauss-Codazzi relations to project the 6D Ricci scalar R_6 onto the 4D hypersurface (with extrinsic curvature tensor $K_{\mu\nu}$), we find:

$$R_6 = R_4 + K^2 - K_{\mu\nu}K^{\mu\nu} + \nabla_\mu(a^\mu)$$

where R_4 is the intrinsic Ricci scalar of our 4D brane and K is the trace of the extrinsic curvature.

F.2 Derivation of 4D Einstein Field Equations

Integrating the action over the brane thickness δ results in the effective 4D Einstein-Hilbert action:

$$S_{eff} = (1 / 2\kappa_4^2) \int d^4x \sqrt{(-g)} R_4 + S_{matter}$$

Varying S_{eff} with respect to the induced 4D metric $g^{\mu\nu}$ yields the classic Einstein Field Equations:

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$$

This demonstrates that gravity is not a quantum field-theoretic phenomenon in BMI, but the macro-geometric response of the 4D hypersurface to bulk deformation.

F.3 Special Relativity and Local Lorentz Invariance

In the limit where the induced metric $g_{\mu\nu}$ is flat ($\eta_{\mu\nu}$), the 4D hypersurface preserves global $SO(3,1)$ symmetry. The Lorentz group $SO(3,1)$ arises as the stabilizer subgroup of the brane's embedding in the 6D bulk, formally recovering Special Relativity as the local geometric baseline for all field dynamics.